

## Sequence & Series Practice

Nine Education IIT Academy | Class 11 Mathematics | 100 Problems: AP, GP, HP, AGP &amp; Square/Cube Series

### Questions

1. The 7th term of an AP is 34 and the 13th term is 64. Find the 20th term.
2. If the sum of first  $n$  terms of an AP is  $S_n = 3n^2 + 5n$ , find its 10th term.
3. Insert 3 arithmetic means between 2 and 18.
4. Find the sum of first 15 terms of an AP with first term 3 and common difference 4.
5. Which term of the AP 3, 8, 13, ... is 78?
6. Find the sum of all natural numbers between 1 and 100 divisible by 5.
7. If  $2k+1$ ,  $3k+5$ ,  $5k+11$  are in AP, find  $k$ .
8. The sum of three numbers in AP is 27 and their product is 504. Find the numbers.
9. Find the sum of first 20 terms of an AP whose  $n$ th term is  $3n+2$ .
10. Is 301 a term of the AP 5, 11, 17, ...?
11. Find the sum of first 25 terms of the AP 7, 10, 13, ...
12. The angles of a triangle are in AP, and the greatest is double the smallest. Find the angles.
13. Find the sum of all 3-digit numbers divisible by 7.
14. If the  $p$ th term of an AP is  $1/q$  and the  $q$ th term is  $1/p$ , find the sum of the first  $pq$  terms.
15. How many terms of the AP 9, 17, 25, ... must be taken to give a sum of 636?
16. Find three numbers in AP whose sum is 15 and product is 105.
17. The sums of  $n$  terms of two APs are in the ratio  $(3n+8):(7n+15)$ . Find the ratio of their 12th terms.
18. Find the middle term of the AP 1, 8, 15, ..., 505.
19. If  $a$ , 7,  $b$ , 23,  $c$  are in AP, find  $a$ ,  $b$ ,  $c$ .
20. Find the sum of the series  $2+5+8+\dots+(3n-1)$  in terms of  $n$ .
21. In a GP, the 4th term is 24 and the 7th term is 192. Find the first term and common ratio.
22. Find the sum of first 6 terms of the GP 2, 6, 18, ...
23. Find the sum to infinity of the GP 8, 4, 2, 1, ...
24. Insert two geometric means between 4 and 32.
25. Find the  $n$ th term of the GP 5, 10, 20, ...
26. The sum to infinity of a GP is 15 and the sum of squares of its terms is 45. Find the first term.
27. Three numbers with sum 21 are in AP; if 2, 3, 9 are added respectively they form a GP. Find the numbers.
28. Find the sum of first 5 terms of a GP with  $a=1$ ,  $r=1/3$ .
29. The 5th term of a GP is 81 and the 2nd term is 3. Find the first term and common ratio.
30. A ball dropped from 100 m rebounds to  $3/4$  of its height each time. Find the total distance travelled before it comes to rest.
31. Find the 6th term of a GP whose first term is 3 and common ratio is  $-2$ .
32. The sum of first three terms of a GP is  $13/3$  and their product is 1. Find the terms.

**33.** If the 1st and nth terms of a GP are a and b, and P is the product of the first n terms, show that  $P^2 = (ab)^n$ .

**34.** Find the sum to infinity of the GP  $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$

**35.** The first term of a GP is 1. The sum of the 3rd and 5th terms is 90. Find the common ratio.

**36.** How many terms of the GP  $3, 3^2, 3^3, \dots$  are needed to give a sum of 120?

**37.** Find a GP whose 2nd term is 6 and 5th term is 162. Write its first 4 terms.

**38.** Find the sum of the series  $0.6 + 0.66 + 0.666 + \dots$  to n terms.

**39.** If the pth term of a GP is q and the qth term is p, find its nth term.

**40.** Three numbers form an increasing GP. If the middle term is doubled, the new numbers are in AP. Find the common ratio.

**41.** Find the 10th term of the HP  $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$

**42.** If the 4th and 9th terms of an HP are  $\frac{1}{10}$  and  $\frac{1}{20}$ , find the 12th term.

**43.** Insert two harmonic means between 3 and 12.

**44.** Find the harmonic mean of 4 and 16.

**45.** If the AM and GM of two numbers are 10 and 8 respectively, find their HM.

**46.** The 3rd term of an HP is 8 and the 6th term is 4. Find the 9th term.

**47.** If the AM and GM of two positive numbers are 13 and 12 respectively, find the numbers.

**48.** The 5th term of an HP is  $\frac{1}{9}$  and the 11th term is  $\frac{1}{21}$ . Find the first term of the HP.

**49.** Find the harmonic mean of 1, 2, 3, and 4.

**50.** The AM and HM of two numbers are 25 and 9 respectively. Find the numbers.

**51.** In an HP, the pth term is q and the qth term is p. Find the (pq)th term.

**52.** Check whether  $\frac{1}{5}, \frac{1}{8}, \frac{1}{11}, \frac{1}{14}$  are in HP.

**53.** Find the nth term of the HP  $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$

**54.** Find the harmonic mean between  $\frac{1}{3}$  and  $\frac{1}{9}$ .

**55.** If 2, x, 8 are in HP, find x.

**56.** Find the sum to infinity of  $1 + 2x + 3x^2 + 4x^3 + \dots$  ( $|x| < 1$ ).

**57.** Find the sum to infinity of  $1 + 4x + 7x^2 + 10x^3 + \dots$

**58.** Find the sum to infinity of  $2 + 6x + 10x^2 + 14x^3 + \dots$

**59.** Find the sum to infinity of  $1 + 3(\frac{1}{2}) + 5(\frac{1}{2})^2 + 7(\frac{1}{2})^3 + \dots$

**60.** Find the sum to infinity of  $3 + 7(\frac{1}{3}) + 11(\frac{1}{3})^2 + 15(\frac{1}{3})^3 + \dots$

**61.** Find the sum to n terms of  $1 + 2x + 3x^2 + \dots + nx^{n-1}$ .

**62.** Find the sum to infinity of  $1 + \frac{3}{4} + \frac{5}{16} + \frac{7}{64} + \dots$

**63.** Find the sum to infinity of  $2 + 5(\frac{1}{5}) + 8(\frac{1}{5})^2 + 11(\frac{1}{5})^3 + \dots$

**64.** Find the sum to infinity of  $1 + 2(\frac{3}{4}) + 3(\frac{3}{4})^2 + 4(\frac{3}{4})^3 + \dots$

**65.** Find the sum to infinity of  $3 + 6(\frac{1}{2}) + 9(\frac{1}{2})^2 + 12(\frac{1}{2})^3 + \dots$

**66.** Find the sum of the first 5 terms of the series  $1 + 3(2) + 5(4) + 7(8) + 9(16)$ .

**67.** Find the 4th term of the series whose nth term is  $(2n-1) \cdot 3^{n-1}$ .

**68.** Find the sum to infinity of  $1 + 2(\frac{1}{3}) + 3(\frac{1}{3})^2 + 4(\frac{1}{3})^3 + \dots$

**69.** Find the sum to infinity of

$$5+9(1/2)+13(1/2)^2+17(1/2)^3+\dots$$

**70.** Find the sum of the first 3 terms of the series

$$1\cdot3+2\cdot3^2+3\cdot3^3+\dots$$

**71.** Find the sum to infinity of

$$4+7(1/4)+10(1/4)^2+13(1/4)^3+\dots$$

**72.** Find the sum to infinity of

$$(1/2)+2(1/2)^2+3(1/2)^3+4(1/2)^4+\dots$$

**73.** Find the sum to infinity of

$$3(1/3)+6(1/3)^2+9(1/3)^3+\dots$$

**74.** Find the sum to infinity of

$$1+5(1/5)+9(1/5)^2+13(1/5)^3+\dots$$

**75.** Find the sum to infinity of

$$2+4(1/2)+6(1/2)^2+8(1/2)^3+\dots$$

**76.** Find the sum of squares of the first 15 natural numbers.

**77.** Find the sum of cubes of the first 10 natural numbers.

**78.** Find the sum of squares of the first 20 natural numbers.

**79.** Find the sum  $1^2+3^2+5^2+\dots+19^2$ .

**80.** Find the sum of cubes of the first 8 even numbers.

**81.** Find the sum of cubes of the first 5 odd numbers.

**82.** Evaluate  $1^3+2^3+3^3+\dots+15^3$ .

**83.** Find the sum of squares of the first 7 odd numbers.

**84.** Find the sum of squares of the first 10 even numbers.

**85.** If the sum of the first  $n$  natural numbers is 210, find  $n$ .

**86.** Find the sum of squares of the first 25 natural numbers.

**87.** Evaluate the sum of  $(r^3-r^2)$  for  $r=1$  to 6.

**88.** Evaluate the sum of  $(2r+1)^2$  for  $r=1$  to 5.

**89.** If the sum of squares of the first  $n$  natural numbers is 2870, find  $n$ .

**90.** If the sum of cubes of the first  $n$  natural numbers is 4356, find  $n$ .

**91.** Evaluate the sum of  $r^2(r+1)$  for  $r=1$  to 10.

**92.** Find the sum of squares of the first 12 natural numbers minus the sum of the first 12 natural numbers.

**93.** Evaluate  $1^3-2^3+3^3-4^3+\dots+9^3-10^3+11^3$ .

**94.** If the sum of squares of the first  $n$  natural numbers is 1015, find  $n$ .

**95.** Evaluate the sum of  $(r^3+3r^2+3r)$  for  $r=1$  to 5.

**96.** Find the sum of the first 12 terms of the series  $1^2+2^2+3^2+\dots$

**97.** Find the sum  $2^3+4^3+6^3+\dots+20^3$ .

**98.** Find  $n$  if  $1^3+2^3+\dots+n^3=225$ .

**99.** Find the sum of squares of the first  $n$  natural numbers in terms of  $n$ , and evaluate for  $n=30$ .

**100.** Evaluate the sum of  $r^3$  for  $r=1$  to 6, plus the sum of  $r^2$  for  $r=1$  to 6.

**Answer Key**

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|------------------------------------|-------------------------------------------------------|-----------------------------|
| 1. 99                              | 35. $r = \pm 3$                                       | 68. $9/4$                   |
| 2. 62                              | 36. 4                                                 | 69. 18                      |
| 3. 6, 10, 14                       | 37. 2, 6, 18, 54                                      | 70. 102                     |
| 4. 465                             | 38. $S_n = (2/3)n - (2/27)(1 - (1/10)^n)$             | 71. $20/3$                  |
| 5. 16th term                       | 39. $T_n = q^{((n-p)/(p-q))} \cdot p^{((q-n)/(p-q))}$ | 72. 2                       |
| 6. 1050                            | 40. $r = 2 + \sqrt{3}$                                | 73. $9/4$                   |
| 7. $k = -2$                        | 41. $1/21$                                            | 74. $5/2$                   |
| 8. 4, 9, 14                        | 42. $1/26$                                            | 75. 8                       |
| 9. 670                             | 43. 4, 6                                              | 76. 1240                    |
| 10. No                             | 44. $32/5$                                            | 77. 3025                    |
| 11. 1075                           | 45. $32/5$                                            | 78. 2870                    |
| 12. $40^\circ, 60^\circ, 80^\circ$ | 46. $8/3$                                             | 79. 1330                    |
| 13. 70336                          | 47. 18, 8                                             | 80. 10368                   |
| 14. $(pq+1)/2$                     | 48. 1                                                 | 81. 1225                    |
| 15. 12                             | 49. $48/25$                                           | 82. 14400                   |
| 16. 3, 5, 7                        | 50. 45, 5                                             | 83. 455                     |
| 17. $7/16$                         | 51. 1                                                 | 84. 1540                    |
| 18. 253                            | 52. Yes                                               | 85. 20                      |
| 19. $a=-1, b=15, c=31$             | 53. $1/(2n-1)$                                        | 86. 5525                    |
| 20. $n(3n+1)/2$                    | 54. $1/6$                                             | 87. 350                     |
| 21. $a=3, r=2$                     | 55. $16/5$                                            | 88. 285                     |
| 22. 728                            | 56. $1/(1-x)^2$                                       | 89. 20                      |
| 23. 16                             | 57. $(1+2x)/(1-x)^2$                                  | 90. 11                      |
| 24. 8, 16                          | 58. $2(1+x)/(1-x)^2$                                  | 91. 3410                    |
| 25. $T_n = 5 \cdot 2^{n-1}$        | 59. 6                                                 | 92. 572                     |
| 26. $a = 5$                        | 60. $15/2$                                            | 93. 756                     |
| 27. 3, 7, 11                       | 61. $[1 - (n+1)x^n + nx^{(n+1)}]/(1-x)^2$             | 94. 14                      |
| 28. $121/81$                       | 62. $20/9$                                            | 95. 435                     |
| 29. $a=1, r=3$                     | 63. $55/16$                                           | 96. 650                     |
| 30. 700 m                          | 64. 16                                                | 97. 24200                   |
| 31. -96                            | 65. 12                                                | 98. 5                       |
| 32. $1/3, 1, 3$                    | 66. 227                                               | 99. $n(n+1)(2n+1)/6$ ; 9455 |
| 33. $P^2 = (ab)^n$                 | 67. 189                                               | 100. 532                    |
| 34. $3/4$                          |                                                       |                             |